Problem Statement — Smart Irrigation Alert Dashboard

1. Introduction

Farmers and agricultural managers face significant challenges in monitoring and managing irrigation across multiple farm zones. Traditional irrigation relies on fixed schedules or manual soil checking, both of which lead to inefficiency — either wasting water through over-irrigation or losing crop yield through under-irrigation.

2. Problem

There is no centralized, real-time system available to small and medium-scale farmers that:

- Collects soil moisture and temperature data from multiple zones
- Displays the data on a searchable dashboard
- Automatically alerts the user when moisture levels fall below a safe threshold
- Provides historical data and summary indicators for informed decision-making

3. Proposed Solution

A **Jenkins-Based Smart Irrigation Alert Dashboard** — a web application that simulates IoT sensor data, displays it on a real-time dashboard, and generates alerts when irrigation is needed. The project will be developed and deployed using a full DevOps pipeline (Git → Jenkins → Selenium → Docker → Ansible).

4. Target Users

#	User	Role
1	Farmers / Farm Owners	View dashboard, receive alerts, act on irrigation recommendations
2	Agricultural Managers / Agronomists	Monitor multiple farm sectors, analyze historical trends
3	System Administrators	Maintain the application, sensors, and infrastructure

5. Existing Pain Points

1. Manual soil moisture checking is time-consuming and error-prone
2. Scheduled irrigation wastes water (no sensor-based decision)
3. No centralized view for multi-zone monitoring
4. Delayed response to weather changes or equipment failures
5. Lack of alert mechanisms for critical moisture levels

6. Stakeholders

- Farmers and Farm Owners (primary users)
- Agronomists and Crop Consultants
- IoT Sensor Providers and Maintenance Teams
- Government Agricultural & Water Conservation Departments

7. Constraints

- **Technical:** Requires internet connectivity for data transmission
- **Budget:** Must use open-source/free tools (Jenkins, Docker, GitHub, Ansible)
- **Time:** MVP must be delivered within the academic semester
- **Hardware:** Sensor data will be simulated (no real IoT hardware required for MVP)

8. Measurable Success Criteria

#	**Criteria**	**Target**
1	Dashboard displays real-time data	At least 3 simulated sensor zones
2	Alert response time	Within 5 seconds of threshold breach
3	System uptime during testing	100%
4	End-to-end deployment time	Under 5 minutes (commit to container)
5	Automated tests	At least 3 Selenium test cases passing

9. Approved MVP Scope

The MVP covers 5 core features:

1. **Data/Event Entry** — Simulate IoT sensor data (moisture levels, temperature)
2. **Searchable Dashboard** — Web UI to view and search sensor statuses
3. **Summary Indicators** — Avg moisture, total zones, active alerts count
4. **Status Drill-Down** — Click on a zone to view detailed readings
5. **Alert/Exception View** — Highlight zones where moisture $< 30\%$ threshold